THE PLANT ARCHITECTURE INVERSION EXPERIMENT

Coherence Gradient Field Specification Produces Pre-Specified Architectural Inversion in *Raphanus sativus* — Operative Protocol v2.3

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The organism is not the output of a genome alone. It is the output of a genome operating within a coherence gradient field. Change the field — the genome executes a different program. Same genome. Different field. Different organism. Placing a radish seed in an inverted coherence gradient field — hydroponic foam above, UV-A embedded foam below, complete external light deprivation — produces root growing upward, shoot growing downward. Derived geometrically. Prediction locked before the experiment. Pre-registered: doi.org/10.5281/zenodo.19081257

The Theorem

Standard: Genome → Organism. **Complete:** Genome + Field → Organism.

Standard architecture (root down, shoot up) is the attractor the genome resolves to in the standard gradient field. **Invert the field before germination: the genome resolves to the inverted attractor.** No genetic modification. One generation.

Root navigates toward water and minerals. **Shoot** navigates toward the only photonic signal in the field — UV-A embedded within the substrate below.

Signal Deprivation Principle

Scenarios 2–5 are in **complete external light deprivation.** The embedded UV-A LED is the *only* photonic signal in the coherence field. The seedling is in **skotomorphogenic state** — maximum photoreceptor sensitivity, zero ambient noise.

A book falling in a silent room is orders of magnitude louder than a shout in a noisy one.

UV-A efficiency relative to peak blue light is irrelevant. Signal-to-noise ratio is effectively infinite. Light exclusion is the load-bearing engineering constraint.

Pre-Registered Prediction — Scenario 3

Root: Upward. Hydrotropism overrides gravitropism. MIZ1-mediated amyloplast degradation suppresses gravity sensing (Zhang et al. PNAS 2025). Water in S1 Type A foam above is the only hydric signal.

Shoot: Downward. UV-A embedded in S2 Type B foam below is the only photonic signal. Shoot penetrates foam to reach it.

AIS = 2 at T=72 h, maintained T=168 h.

AIS = 2: root >135°, shoot <45°from vertical. Scenario 1 must record AIS = 0 in same run. Minimum 3 seeds, 2 independent runs for reportable confirmation.

Confirmed Literature Support

Root upward to water, radish	Takahashi 2003
MIZ1 gravity suppression, drought	Zhang PNAS 2025
Hydrotropism > gravitropism	PNAS 2025, 1994
Shoot bends toward buried light	Phototropin lit.
Root +ve phototropism, no gravity	Kiss ISS 2012
Photosynthesis from buried geothermal light	Beatty PNAS 2005
Radiotropism toward radiation source	Dadachova 2007

No mechanism is novel in isolation. The combination, geometric pre-specification, and simultaneous full architectural inversion from a locked prediction are novel. Every prior result is post-hoc or partial.

The Five Scenarios — v2.3

Substrate types: Free UV — LED free in S1 air, zero penetration cost. **Type A** — 20 ppi foam + hydroponic (water + minerals, no UV). **Type B** — 20 ppi foam + UV-A LED embedded (UV only, no hydroponic). **Type C** — 20 ppi foam + hydroponic + UV-A LED embedded (all co-located).

ID	S1 top	S2 bottom	Cone
S1	Free UV in air	Type A	Apex up
S2	Type B (UV only)	Type A	Apex up
S3	**Type A**	**Type B (UV only)**	**Apex dn**
S4	Type C (all)	Empty void	Apex up
S5	Empty void	Type C (all)	Apex dn

Cone rule: apex always toward void or least active substrate.

S3 is the primary claim. S1 is the validity gate. S2 vs S3 is the primary inversion pair. S4 and S5 are single-polarity maximum field tests.

What Confirmation Establishes

1. Field authorship of architecture. Same genome, inverted field, inverted organism. Architecture is a field output. First demonstration from a pre-specified locked prediction.

2. Attractor geometry is real and navigable. The field is the instruction set. The genome is the build capacity. The organism is the output of both.

3. No genome editing required. Specify the field. The genome executes.

Materials and Timeline

Item	Cost	Source
Radish seeds	\$2–4	Garden store
Reticulated foam 20 ppi	\$5–10	Aquarium supply
Hydroponic mineral solution	\$5–12	Garden / hydro
Opaque PVC pipe + end caps	\$15–25	Hardware store
UV-A LED strip ∼365 nm	\$8–15	Amazon
Metallized Mylar + funnels + net cups	\$5–11	Dollar / kitchen
Aquarium silicone sealant	\$5–8	Aquarium store
Red-light torch	\$5–15	Astronomy supply
Total	**\$50–100**	

No BSL rating. No credentials. No specialist equipment.

Light exclusion is the critical build requirement.

Primary result T = 72 h. Full confirmation T = 168 h.

The field came first. The genome was always listening. The architecture was never fixed. The inversion was always possible.

github.com/Eric-Robert-Lawson/attractor-oncology · doi.org/10.5281/zenodo.19081257 · ORCID: 0009-0002-0414-6544